

Personal eVTOL Hover-Vehicle

An Innovative All-Terrain Personal Transport System Utilizing Advanced Autonomous Flight Stabilization Control & Frictionless Ground Effect Physics.

RUET IDEAS PROGRAMME 2026 • DEPARTMENT OF ECE

| The Vision: **Frictionless Mobility**

 Futuristic eVTOL Hoverbike personal flight vehicle in motion

Why Eliminate Wheels?

Traditional transportation gets stuck on complex soil profiles. By flying 1 to 2 feet above the soil, we eliminate mechanical rolling resistance entirely.

- ✓ **Zero-Infrastructure Dependencies:** Traverses mud, sand, and water.
- ✓ **Eco-Friendly Architecture:** Fully electric, producing zero carbon emissions.
- ✓ **Ground Effect Boost:** Uses 30% less energy than high-altitude drones.

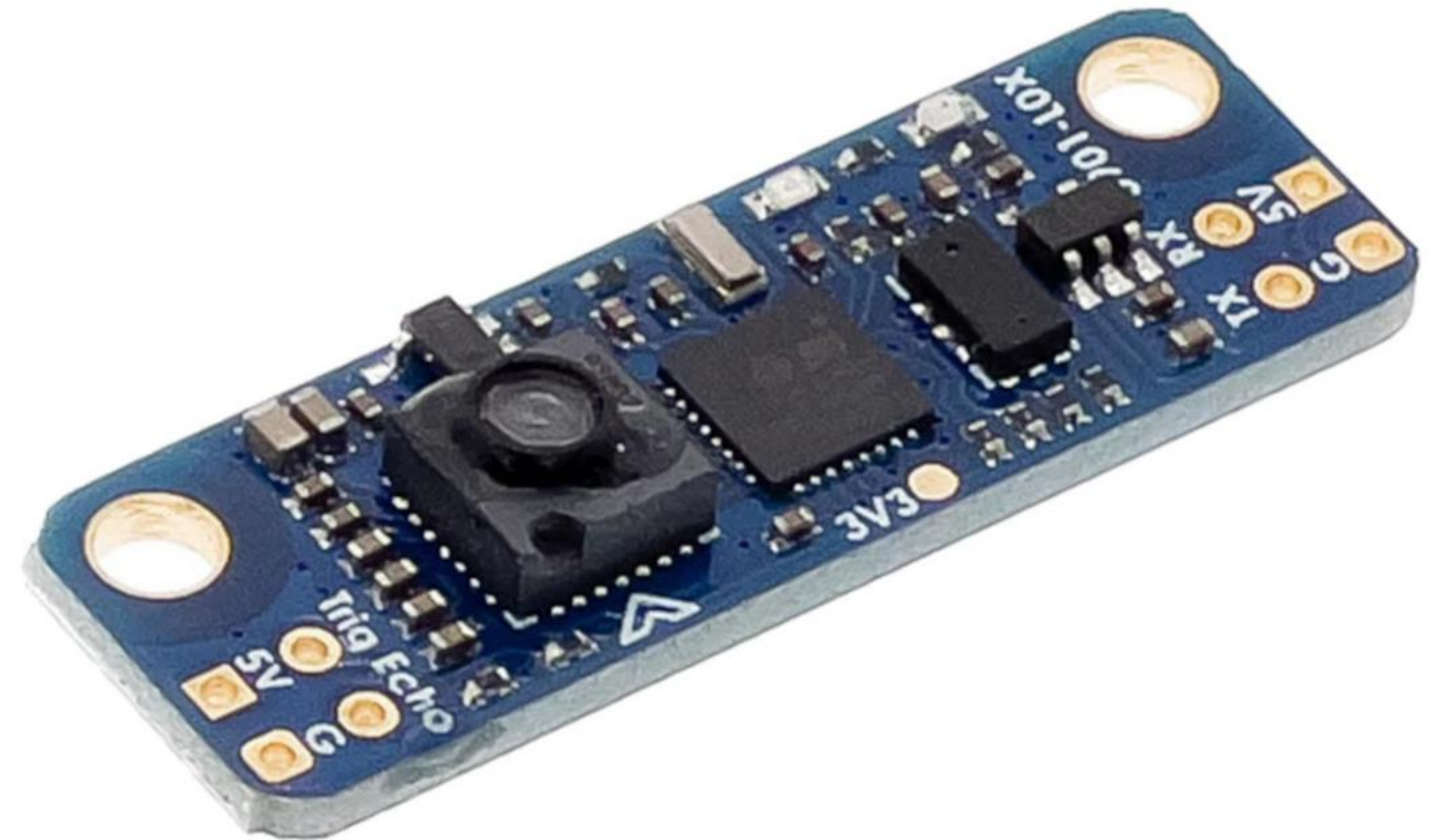
ECE Architecture: **Active Stabilization**

Autonomous Flight Controller

Maintaining safety near the soil requires high-speed sensor feedback and microsecond adjustments. The pilot leans, and the flight computer automatically calculates stabilization matrix commands.

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

Dynamic PID controller algorithm adjusting PWM output to multi-rotor ESCs based on IMU tilt errors.



| Target Market: **Four Key Sectors**



Defense & Patrol

Enabling border forces to patrol rugged terrains like marshes, border rivers, and coastal mudbanks securely where standard vehicles cannot drive.



Disaster Relief

Designed for search & rescue teams to traverse flash-flooded zones, debris field obstacles, and devastated infrastructure instantly during critical response windows.



Smart Agriculture

Large-scale precision crop health surveillance and targeted automated liquid dispersal without risking soil compaction or physical crop destruction.

| Sustainable **Business Model**



Revenue Streams

Divided into two direct pipelines: direct manufacturing sales (B2G/B2B procurement) and continuous SaaS revenue via proprietary flight software updates.



Cost Distribution

Optimized through modular assembly. Heavy upfront investments in sensor array calibration (LiDAR/IMU) and carbon-composite molds, reducing long-term variable labor costs.



Strategic Channels

Acquiring industrial validation by working alongside government disaster authorities, research institutions, and large agricultural cooperatives via pilot demonstration leases.

Comparative Analysis: How We Outperform

Operational Factor	Standard Drone (eVTOL)	Mechanical Hovercraft	Our stabilized Hover-Vehicle
Energy Consumption	High (Constantly fights gravity)	Moderate (Air cushion lift)	Low (Ground Effect Physics)
Steering & Agility	Highly Agile	Difficult (Drifts over terrain)	Highly Precise (PID Vectoring)
Noise Levels	Very High Propeller Pitch	Deafening Fan Engine	Low (Multi-ducted ultra-quiet fans)
Terrain Versatility	Airbound Only	Flat surfaces & water	All rugged soil & water profiles

| Fail-Safe Reliability Architecture

Dual

MCU REDUNDANCY SETUP

Multi-tier Safe Landing Routine

Operating close to the soil demands robust fail-safe protocols. In case of primary sensor failure, the system behaves autonomously:

- Secondary MCU takes over in under **2 milliseconds** if primary crashes.
- Dual down-facing Ultrasonic sensors act as backup to the primary LiDAR array.
- Propellers are fully shrouded in safety ducts, protecting the rider entirely.

Join the Revolution

"The best way to predict the future is to design it." Let's build the future of all-terrain personal transit together.

THANK YOU • OPEN FOR QUESTIONS

Presented by: ECE Department, Rajshahi University of Engineering & Technology (RUET)

| Image Sources



<https://futurism.com/wp-content/uploads/2022/01/inventor-builds-real-life-personal-hover-bike.jpg?quality=85&w=1152>

Source: futurism.com



https://probots.co.in/pub/media/catalog/product/cache/d8ddd0f9b0cd008b57085cd218b48832/m/a/matek_3901-l0x_optical_flow_sensor_module_5_.jpg

Source: probots.co.in